Program – 2

Develop a Multi-Stage Dockerfile for Container Orchestration

Program Structure

program2/
------>node_modules
------>package.json
------>package-lock.json
------->build|
--->src
----->index.js
------>dist
------->index.js
-------->Dockerfile

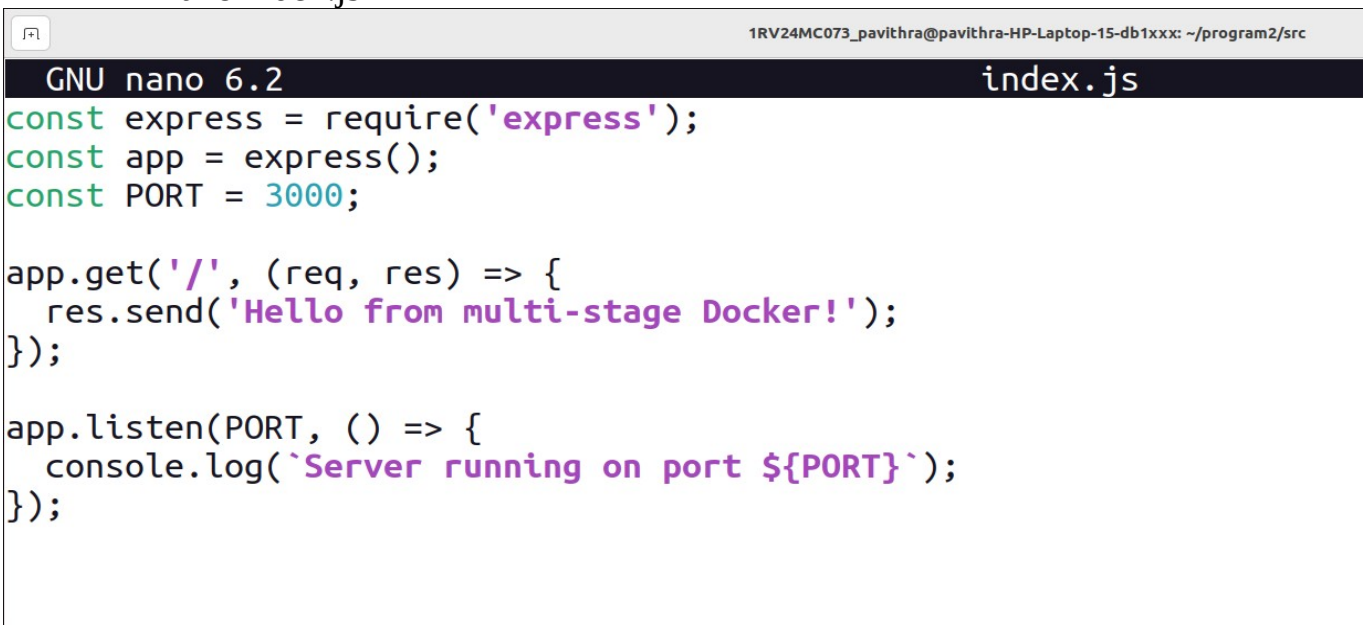
Step 1: Create a project folder
mkdir program2
cd program2
Step 2 : initialize npm – it will create package.json
npm init -y

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~$ mkdir program2
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~$ cd program2
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program2$ nano Dockerfile
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program2$ mkdir src
nano src/index.js
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program2$ nano package.json
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program2$ npm init -y
Wrote to /home/1RV24MC073_pavithra/program2/package.json:

{
  "name": "program2",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

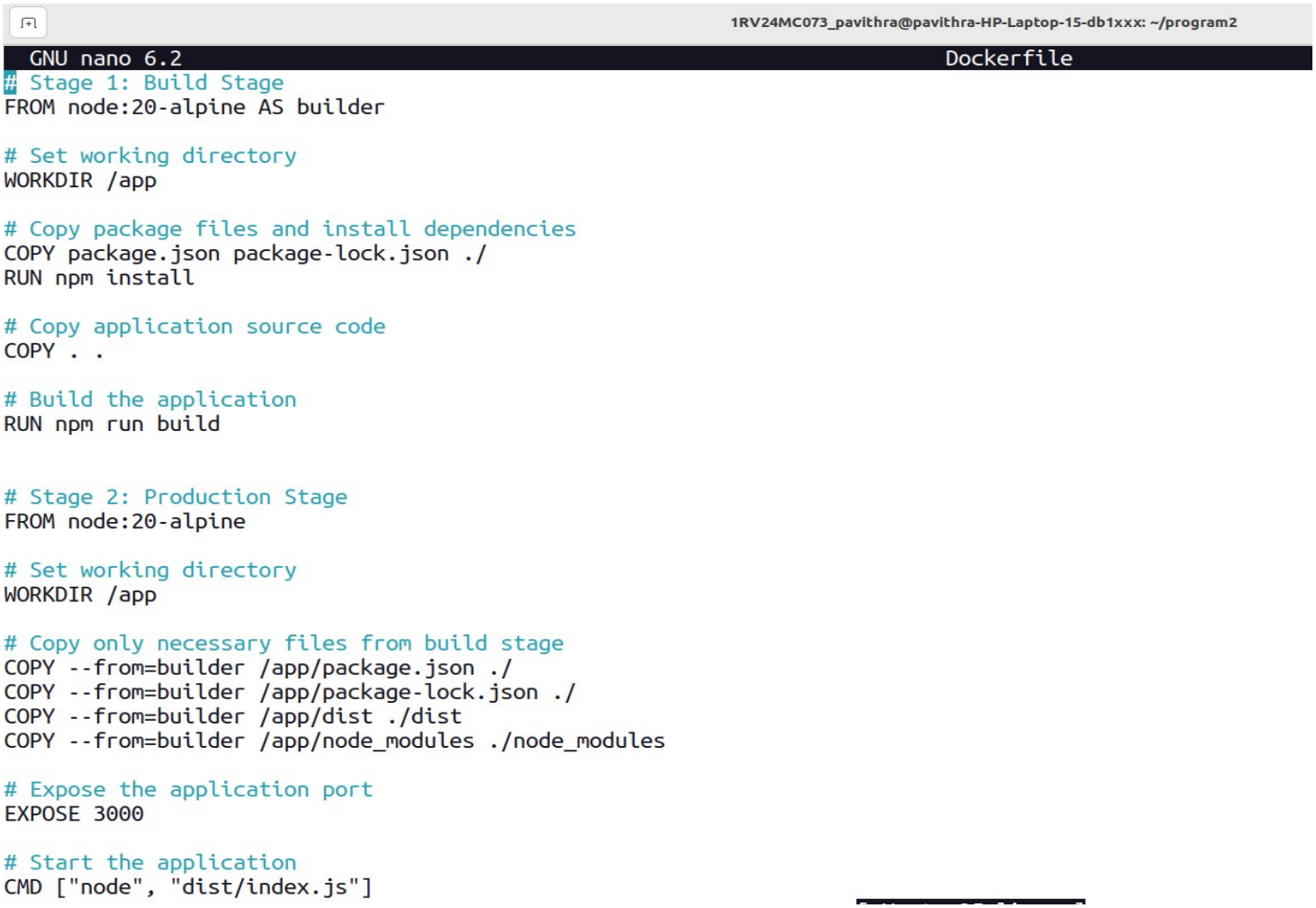
Step 3 : create a src folder and create index.js file inside this folder
mkdir src
cd src

nano index.js



Step 4 : create a docker file

nano Dockerfile



Step 5 : build docker image
docker build -t program2 .
run the container
docker run -p 3000:3000 program2

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/program2
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program2$ nano package.json
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program2$ npm run build

> program-2@1.0.0 build
> mkdir -p dist && cp -r src/* dist/

1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program2$ docker build -t program2 .
[+] Building 15.2s (16/16) FINISHED                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                          0.0s
 => => transferring dockerfile: 792B                                                                          0.0s
 => [internal] load metadata for docker.io/library/node:20-alpine                                             2.1s
 => [auth] library/node:pull token for registry-1.docker.io                                                   0.0s
 => [internal] load .dockerignore                                                                             0.0s
 => => transferring context: 2B                                                                               0.0s
 => [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:658d0f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488  0.0s
 => [internal] load build context                                                                             0.0s
 => => transferring context: 811B                                                                             0.0s
 => CACHED [builder 2/6] WORKDIR /app                                                                         0.0s
 => [builder 3/6] COPY package.json package-lock.json ./                                                      0.1s
 => [builder 4/6] RUN npm install                                                                            11.3s
 => [builder 5/6] COPY . .                                                                                    0.1s
 => [builder 6/6] RUN npm run build                                                                           0.7s
 => [stage-1 3/6] COPY --from=builder /app/package.json ./                                                    0.1s
 => [stage-1 4/6] COPY --from=builder /app/package-lock.json ./                                               0.1s
 => [stage-1 5/6] COPY --from=builder /app/dist ./dist                                                        0.1s
 => [stage-1 6/6] COPY --from=builder /app/node_modules ./node_modules                                        0.2s
 => exporting to image                                                                                        0.2s
 => => exporting layers                                                                                       0.1s
 => => writing image sha256:40a89f497c951f3e77c2cd51956ade09875c412797f55af53a673d14d76dc3f7                  0.0s
 => => naming to docker.io/library/program2                                                                   0.0s
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program2$ docker run -it -p 3000:3000 program2
Server running on port 3000
```

Step 7 : verify the application
open the browser and go to http://localhost:3000